



A Communication Effectiveness Study of the FDA GMP Warning Letters

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Abstract

Purpose Study the structure and content trend over time of the FDA GMP Warning Letters for their roles in effective communication.

Methods The structure and content trend over time of Warning Letters are studied using a combination of parsing text based on function, manual coloring and Python automation.

Results The observed trends using these methods indicate that the content of suggestions on how to respond to a Warning Letter, for third party participation and further reading consistently increases over time.

Conclusions Warning Letters have become increasingly informative, showing FDA's good practice of transparency as a way to improve regulatory effectiveness.

Keywords Warning letters · Word count · Communication effectiveness · Transparency

Introduction

Transparency has recently been suggested by a theoretical modeling work to be fundamentally important to regulatory science and regulatory effectiveness, and the importance can be quantitatively estimated [1]. The present work intends to identify and quantify practical evidence that may suggest transparency has indeed been used to improve regulatory effectiveness.

Pharmaceutical manufacturing is required to follow the Good Manufacturing Practice (GMP) regulation by the Food and Drug Administration (FDA) in the United States [2]. Manufacturing firms not in compliance with the GMP regulation run the risk of making ineffective or unsafe products, and of facing regulatory actions. The FDA's GMP program

has two major components. One is the written regulation, along with related guidances and policies, which are a rich set of dos and don'ts along with rationales [3]. Another is enforcing the regulation by inspecting manufacturing firms and taking regulatory actions for GMP violation [4].

A major consequence for GMP violation is an FDA Warning Letter, which puts a firm's drug application approval on hold, and warns the firm that further enforcement actions may follow, unless its GMP compliance is substantially and immediately improved [5]. A published Warning Letter also puts the firm's partners on notice about the risk of doing business with the firm.

Many factors can contribute to a compliance failure, among them is lack of understanding of the regulation. A Warning Letter also clarifies regulatory requirements in the context of compliance practice. For example, a Warning Letter often states what the GMP requirements are, where the firm meets and fails to meet the requirements, including both facts and reasoning, and what needs to be done to correct deficiencies.

The FDA regulates thousands of firms worldwide, consuming time and resources. It is essential for the regulation to be effective. A good practice of transparency requires not only disclosing information, but also effective communication. Improving regulatory effectiveness by improving the writing of Warning Letters is a good practice of regulatory transparency.

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The FDA Warning Letters have been studied for policy implementation and enforcement priorities [6–8]. The present work focuses on the structure and content trend over time of Warning Letters for their roles in effective communication, along with the regulatory guidances and policy documents available for Warning Letters to reference. Such a structural understanding, in conjunction with the theoretical modeling work [1], may help to measure and improve the communication effectiveness of the future Warning Letters in a more pointed and predicted way.

Methodology

Scholars have analyzed measurable features of the U.S. Supreme Court's written opinions as a means of understanding the dynamics of the Court's decision making process, and suggested that the language of judicial opinions represents the law [9]. The FDA has full authority in making, interpreting and enforcing regulation as authorized by the Food, Drug and Cosmetic Act (FDCA). In this regard, the language used in Warning Letters that are deliberate and formal responses to compliance violations, should represent the compliance enforcement part of the GMP regulation. For this reason, it may be helpful to study the structure and content trend over time of the Warning Letter language as a way to learn about the FDA's practice of transparency to improve communication effectiveness of the GMP regulation.

Data Sources – Warning Letters

The FDA started to publish Warning Letters in 1996, which can be accessed on the following websites as listed in Table 1 through December 31, 2022. Please note a Warning Letter's issuance date may not be the same as its posted date. For the present research, the cut-off issuance date is December 31, 2022, and the cut-off posted date is August 30, 2023. For example, a Warning Letter issued after December 31, 2022 but posted before August 30, 2023 is excluded, and a Warning Letter issued before December 31, 2022 but posted after August 30, 2023 is also excluded.

The present work is only concerned with the human drug GMP (21 CFR 210 and 211, Code of Federal Regulations) related Warning Letters. Drug substances, while not

directly regulated under 21 CFR 210 and 211, are included. Warning Letters issued by Center for Biologics Evaluation and Research, Center for Devices and Radiological Health, Center for Veterinary Medicine, Center for Food Safety and Applied Nutrition, Center for Tobacco Products are excluded. Warning Letters from Center for Drug Evaluation and Research (CDER)'s Office of Prescription Drug Promotion are excluded. Warning Letters for compounding pharmacies, medical gases and hand sanitizers, biological products regulated under 21 CFR 600–680, Positron Emission Tomography drugs regulated under 21 CFR 212 are also excluded. In addition to the above general exclusions, there are two specific exclusions: “1997-02-18_Dr. Mann Pharma” [16], due to its poor Portable Document Format (PDF) quality and lack of technical content (redacted), and “2001-09-27_Unilever HPC-USA” [17], due to its invalid web-link. After these exclusions, out of a total 18,329 of Warning Letters, 1151 are included for study in present work.

Warning Letters from 1996 to the first half of 2001 are in the PDF format. They are first converted to the Microsoft Word format by using an Optical Character Recognition (OCR) software, followed by manual proofreading. Warning Letters from the second half of 2001 to 2022 are in the text format, and are directly converted to the Microsoft Word format. All subsequent analyses of Warning Letters are based on the Microsoft Word format.

Data Sources – GMP Guidances and Policy Documents

The FDA publishes a large number of pharmaceutical GMP guidances and policy documents to help the industry understand how it views and enforces the GMP regulation, of which the following list is covered in present work.

- FDA Guidances.
- ICH (International Council for Harmonisation) Guidelines.
- CPG: Manual of Compliance Policy Guides.
- CPM: Compliance Program Manual.
- FMD: Field Management Directives.
- IG: Inspection Guides.
- IOM: Investigations Operations Manual.
- ITG: Inspection Technical Guides.
- MAPP: CDER Manual of Policies and Procedures.
- RPM: Regulatory Procedures Manual.
- White Papers.
- Withdrawn: Withdrawn | Expired Guidances.

White papers include *Pharmaceutical CGMPs for the 21st Century – A Risk-Based Approach* [18], *Pharmaceutical*

Table 1 Warning Letters and their web-links

Time Period	Website
1996–2004	[10]
2005–2011	[11]
2012	[12]
2013–2014	[13]
2015–2017	[14]
2018–2022	[15]

Quality for the 21st Century – A Risk-Based Approach Progress Report [19], and CDER's *Quality Management Maturity (QMM) Program: Practice Areas and Prototype Assessment Protocol Development* [20]. CGMP stands for current GMP, where letter C emphasizes that the pharmaceutical industry is required to use technologies and systems that are up-to-date in order to comply with the GMP regulation.

A guidance or policy document can undergo several drafts before it is finalized. Unfortunately, the FDA does not provide public access to all draft versions, thus it is often difficult to find the date of the first draft. This date is important, because once a draft guidance is published, the industry usually takes immediate notice and starts to follow. For the purpose of the present work, which is to find the trend over time of the number of available guidances and policy documents as a measure of FDA's effective communication with the industry, the earliest publication date is taken as the earliest known date of a draft or the finalized version. The latest publication date is the date of the current draft or of the finalized guidance or policy document. To find the earliest publication dates, the present work relies on the FDA's website, the Federal Register and a commercial database [21].

Guidances and policy documents from FDA centers other than CDER are included as long as they contain a pharmaceutical GMP part (excluding drug compounding). Guidances and policy documents with an earliest publication date past December 31, 2022 are excluded. Guidances published during the Covid-19 time from 2020 to 2022, but withdrawn in 2023 are included.

Parsing the Content of Warning Letters

The content of each Warning Letter is parsed into nine categories as listed in Table 2.

An FDA Form 483 is issued to a manufacturing firm at the conclusion of a GMP inspection, detailing the observed GMP violations by an FDA investigator(s). The firm is required to respond to Form 483 in writing in fifteen business days. Parsing is performed manually on a sentence by

sentence basis on a Microsoft Word file. Color selections are arbitrary, as long as their Red Green Blue (RGB) codes are distinct from one another.

Content Coloring

Categories 1 and 9 are administrative in nature, and thus are excluded from further analysis. Category 7 is where the FDA suggests a firm seek a 3rd party for assistance. Please note that the FDA does not suggest any particular consultant. While Category 7 is also administrative in nature, its language is pointed and timely, for example (*italic* is assigned by the present work), “The qualified consultant should also perform a comprehensive audit of your *entire* operation for CGMP compliance and evaluate the completion and efficacy of *all* identified corrective actions and preventive actions *before* you pursue resolution of your firm's compliance status with FDA”, and “Your use of a consultant *does not relieve your firm's obligation* to comply with CGMP. Your firm's *executive management remains responsible* for resolving *all* deficiencies and systemic flaws to ensure ongoing CGMP compliance”. Consequently, Category 7 is included for further analysis.

Word Count

A published Warning Letter may have many words redacted to protect a firm's trade secret. The present work only counts the un-redacted words. In general, the amount of redacted words is small, which can be seen from Warning Letters dated prior to 2001. These older Warning Letters were scanned into the PDF format, and the redacted words are blacked out. The visible blacked out spaces show they represent a small portion of a Warning Letter. For newer Warning Letters, a redaction is denoted as (b) (4), which does not represent a blacked out space. However, if one can assume the redaction has been consistent throughout the years, the redacted spaces should consistently represent a small portion of a Warning Letter.

A Python program is written to read and parse a color-marked Warning Letter in the Microsoft Word format (.docx), and to output the word count for each of the nine colors. Such an automatic word count program is helpful for processing over a thousand Warning Letters, especially as the word count may need to be repeated from time to time. This is because the manual reading and coloring of the large number of Warning Letters are error prone, and corrections are often needed. For example, the FDA's Comments (green) to Firm's 483 Response (light green) sometimes are intertwined with the Instruction for Further Response (purple), or Facts of GMP Violation

Table 2 Parsing and coloring of the Warning Letter content

#	Category	Color
1	Introduction	Light Gray
2	CFR and ICH Citation	Red
3	Facts of GMP Violation	Blue
4	Firm's 483 Response	Light Green
5	Comments on Firm's 483 Response	Green
6	Instruction for Further Response	Purple
7	Suggestion for Seeking 3rd Party Help	Light Blue
8	Recommendation for Reading	Orange
9	Conclusion	Gray

Column 1	Column 2	Column 1	Column 2
A	B	A	B
	C	A	C

(a)

Column 1	Column 2
A	B
A	C

(b)

Fig. 1 A merged cell in a table is split, when read by Python

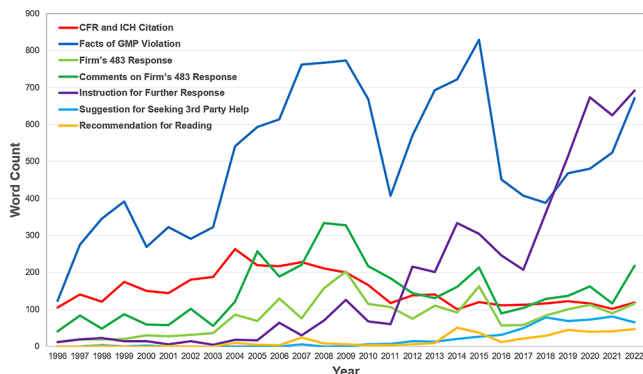


Fig. 2 The average word count per year over time for each category of Warning Letters

(blue), a slight negligence will result in incorrect word counts.

There are three basic concepts in using Python to analyze a Word file, namely, Document, Paragraph and Run. A Document object corresponds to the file. A Paragraph object corresponds to a natural paragraph. A Run object corresponds to a part of a paragraph, which identifies the same style of text continuation, such as font, color and size. Here are three typical cases that need special attention, because invisible characters are involved.

- After a Warning Letter is converted from the PDF to the Word format with an OCR software, there are usually errors to be corrected. For example, word “good” may be mistakenly converted to “goad” or “god”, which can be easily corrected by hand in a Word file. But such a correction will add an invisible character to the underlying Word file, which will break a run object into two, and falsely adds an extra word, when processed by Python.
- While Warning Letters are mostly in the text format, some include tables. To process tables, one needs to use Python’s Document – Table – Row/Column – Cell setup, and then applies Python’s Paragraph – Run program to each cell to get the correct word count.
- When reading a table with a merged cell, as shown in Fig. 1, Python automatically converts the Table (a) to Table (b), and falsely adds an extra cell of A.

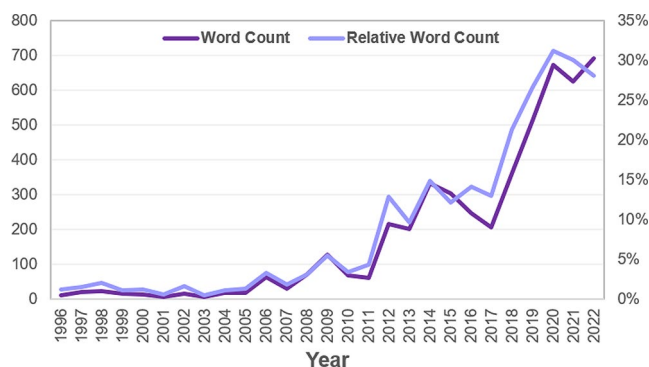


Fig. 3 The absolute and relative average word counts of Instruction for Further Response

Results

The average word count over time per Warning Letter for each of the seven categories (2 to 8) is shown in Fig. 2. Categories of CFR and ICH Citation (red), Facts of GMP Violation (blue), Firm’s 483 Response (light green) and Comments on Firm’s 483 Response (green) show no clear trend over time, especially in recent years. The remaining three categories appear to increase over time, and are examined in detail as follows.

The category that increases most significantly over time is *Instruction for Further Response*, both in terms of the absolute (purple) and relative (light purple) average word count as the percentage of the total word count per Warning Letter as shown in Fig. 3. By 2020, this category has risen to approximately 30% of the space of a Warning Letter from 1–2% two decades ago. To a firm, this category is probably most important, as nothing is more urgent than closing out a Warning Letter as soon as possible, and nothing is more helpful than knowing FDA’s expectation for the firm’s response to the Warning Letter. To the FDA, it is a priority to help the firm to effectively improve GMP compliance to ensure product quality and avoid possible drug shortage.

The category of *Suggestion for Seeking 3rd Party Help* also increases over time significantly. A related indicator, which is the percentage of Warning Letters that suggest a firm to seek 3rd party help, also increases significantly over time, as shown in Fig. 4. This percentage has risen to nearly 90% of Warning Letters from less than 10% twenty years ago. A 3rd party involvement is particularly important for cases involving data integrity issues since 2012. When the integrity of a firm is in doubt, a reputable 3rd party involvement is almost always required.

In a Warning Letter, the FDA may suggest that the firm read guidances and policy documents. The word count for *Recommendation for Reading* is rather uncharacteristic, as a Warning Letter sometimes simply recommends

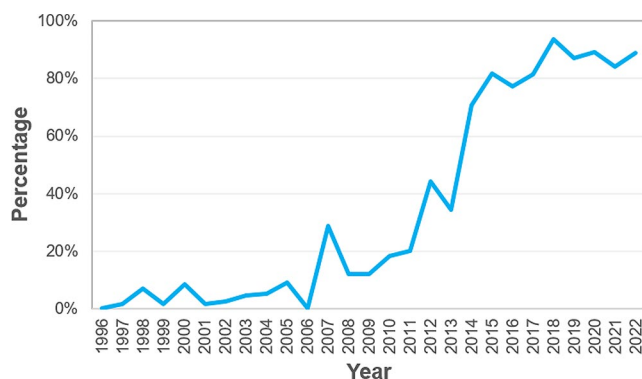


Fig. 4 The percentage of Warning Letters that Recommend Seeking 3rd Party Help

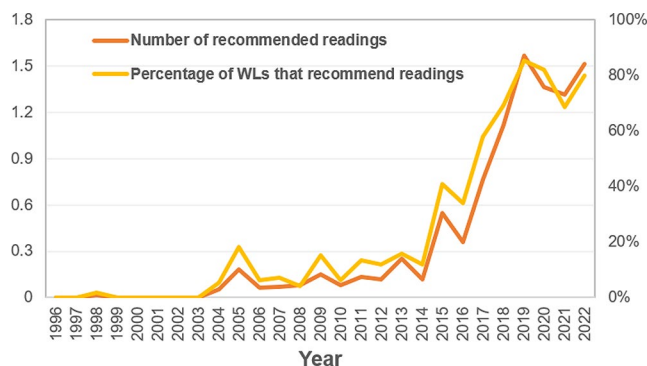


Fig. 5 The number of recommended readings per year in a Warning letter, and the percentage of Warning Letters with such a recommendation

what to read, while other times gives a lengthy introduction. The number of the recommended readings per Warning Letter is a better indicator, and increases significantly over time, as shown in Fig. 5. A related indicator is the percentage of Warning Letters that recommend readings, which also increases significantly over time. Twenty years ago, a Warning Letter hardly recommended any reading, but now nearly 80% do.

The increasing number of recommended readings is probably due to the increase of GMP related guidances and policy documents available for Warning Letters to reference, as shown in Fig. 6. While the increase in guidances and policy documents each year has its ups and downs, the number of cumulated guidances and policy documents increases almost linearly, rising nearly 50% over the past twenty years. This increase may suggest that FDA has become more specific about what is required, as a firm's response to Warning Letters may involve increasingly more people, such as lawyers, who may not be technically well versed.

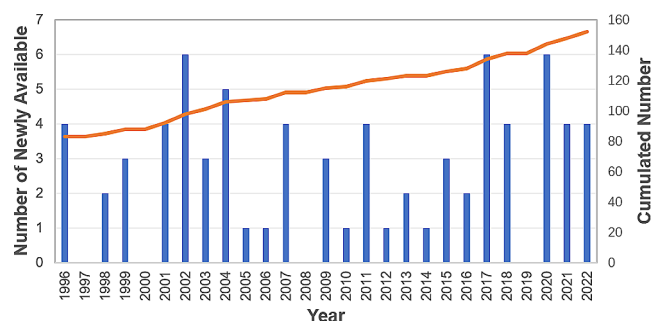


Fig. 6 The number of newly available GMP related guidances and policy documents each year, and the cumulated number of available GMP related guidances and policy documents

Discussion and Conclusion

The word count method used in present work is based on a personal reading of 1151 Warning Letters of more than 1.7 million words, and parsing them, on a sentence by sentence basis. This process is labor intensive, and requires understanding of the GMP regulation. A judgement is often needed to make a parsing reasonable. There are too many such judgements to list for a normal length academic paper, and a thorough evaluation of them is not practical. All these contribute to uncertainties of the results in present work. The automatic Python program runs on top of these uncertainties, and is used only to save time to process, not to reduce uncertainties.

While the structure of Warning Letters has become more consistent over the past twenty years, they are written by people with different writing styles, under different circumstances, and for different compliance violations. All these add to the inherent variations to the content of Warning Letters.

With all the uncertainties of the parsing methodology and the variations within Warning Letters, it is no surprise to see the lack of clear trends over time in Fig. 2. However, upon closer examination, it is remarkable to see the significant increases over time in Figs. 3, 4 and 5. Admittedly, the present work has no quantitative estimate of the signal to noise ratios of these increases. But based on the extensive reading and parsing, re-reading and re-parsing exercises, the present work believes there are more signals than noise in these increases. These signals suggest Warning Letters have become more informative and comprehensive to help firms to more effectively understand and correct GMP violations, and to communicate with the FDA. Specifically, the FDA has become more explicit and specific on how to respond to Warning Letters (Fig. 3), what values a 3rd party consultancy can bring (Fig. 4), what to read to understand regulatory requirement (Fig. 5), and for the broad education to the industry on guidances and regulatory policies (Fig. 6).

With the recent advances in developing large language models (LLMs) [22], the labor intensive approach used in present work can be undoubtedly improved, for example, by configuring ChatGPT in Python to read and analyze Warning Letters, as suggested by a reviewer of the present work. One challenge for the successful use of LLMs is doing a specific task exactly right, as opposed to doing a variety of tasks approximately right. One approach to meet the challenge is through case studies to identify specific surprises, like the hidden characters causing incorrect Python reads, as discussed in present work.

A good practice of transparency requires effective communication. Making Warning Letters more informative is a good practice of transparency, resulting in more effective regulation. In this regard, the present work has identified practical and quantitative evidence that transparency may have indeed been used by the FDA in writing Warning Letters to improve regulatory effectiveness.

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Declarations

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References

1. Bao Y, Buhay N, Zheng Q. A Dynamic Model for GMP Compliance and Regulatory Science. *J Pharm Innov.* 2024. <https://doi.org/10.1007/s12247-024-09825-x>.
2. U.S. Food and Drug Administration. Current Good Manufacturing Practice (CGMP) Regulations. <https://www.fda.gov/drugs/pharmaceutical-quality-resources/current-good-manufacturing-practice-cgmp-regulations>. Accessed 04 Apr 2024.
3. U.S. Food and Drug Administration. Pharmaceutical Quality Documents. <https://www.fda.gov/drugs/pharmaceutical-quality-resources/search-pharmaceutical-quality-documents>. Accessed 04 Apr 2024.
4. U.S. Food and Drug Administration. Warning Letters and Notice of Violation Letters to Pharmaceutical Companies. <https://www.fda.gov/drugs/enforcement-activities-fda/warning-letters-and-notice-violation-letters-pharmaceutical-companies>. Accessed 04 Apr 2024.
5. U.S. Food and Drug Administration. Regulatory Procedures Manual - Warning Letters. <https://www.fda.gov/media/71878/download>. Accessed 04 Apr 2024.
6. Nguyen D, Seoane-Vazquez E, Rodriguez-Monguio R, et al. Changes in FDA enforcement activities following changes in federal administration: the case of regulatory letters released to pharmaceutical companies. *BMC Health Serv Res.* 2013. <https://doi.org/10.1186/1472-6963-13-27>.
7. Wang L, Zheng H, Ren X, et al. Trend analysis of FDA warning letters issued to medical products about violations to current good manufacturing practices (cGMP) between 2007 and 2014. *Therapeutic Innovation & Regulatory Science.* 2016. <https://doi.org/10.1177/2168479015619201>.
8. Rathore A S, Li Y, Chhabra H, et al. FDA warning letters: a retrospective analysis of letters issued to pharmaceutical companies from 2010–2020. *J Pharm Innov.* 2023. <https://doi.org/10.1007/s12247-022-09678-2>.
9. Black R C, Spriggs J F. An empirical analysis of the length of US Supreme Court opinions. *Hous. L. Rev.*, 2008, 45: 621.
10. U.S. Food and Drug Administration. 1996–2004 Archived Warning Letters. <https://web.archive.org/web/20141013125228/http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/2004/default.htm>. Accessed 30 Aug 2023.
11. U.S. Food and Drug Administration. 2005–2011 Archived Warning Letters. <https://wayback.archive-it.org/7993/20170110185432/http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/default.htm>. Accessed 30 Aug 2023.
12. U.S. Food and Drug Administration. 2012 Archived Warning Letters. <https://wayback.archive-it.org/7993/20171113211251/http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/default.htm>. Accessed 30 Aug 2023.
13. U.S. Food and Drug Administration. 2013–2014 Archived Warning Letters. <https://wayback.archive-it.org/7993/20190422112825/http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/default.htm>. Accessed 30 Aug 2023.
14. U.S. Food and Drug Administration. 2015–2017 Archived Warning Letters. <https://wayback.archive-it.org/7993/20201217171658/http://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/compliance-actions-and-activities/warning-letters>. Accessed 30 Aug 2023.
15. U.S. Food and Drug Administration. Warning Letters. <https://www.fda.gov/inspections-compliance-enforcement-and-criminal-investigations/compliance-actions-and-activities/warning-letters>. Accessed 30 Aug 2023.
16. U.S. Food and Drug Administration. Warning Letters of Dr. Mann Pharma issued on 1997-02-18. <https://web.archive.org/web/20120926153445/http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/1997/ucm156131.htm>. Accessed 30 Aug 2023.
17. U.S. Food and Drug Administration. Warning Letters of Unilever HPC-USA issued on 2001-09-27. <https://web.archive.org/web/20130210062309/http://www.fda.gov/ICECI/EnforcementActions/WarningLetters/ucm178319.htm>. Accessed 30 Aug 2023.
18. U.S. Food and Drug Administration. Pharmaceutical CGMPs for the 21st Century – A Risk-Based Approach. 2004. <https://www.fda.gov/media/77391/download>. Accessed 30 Aug 2023.

19. U.S. Food and Drug Administration. Pharmaceutical Quality for the 21st Century – A Risk-Based Approach Progress Report. 2007. <https://www.fda.gov/about-fda/center-drug-evaluation-and-research-cder/pharmaceutical-quality-21st-century-risk-based-approach-progress-report>. Accessed 30 Aug 2023.
20. U.S. Food and Drug Administration. CDER's Quality Management Maturity (QMM) Program: Practice Areas and Prototype Assessment Protocol Development. 2023. <https://www.fda.gov/media/171705/download?attachment>. Accessed 30 Aug 2023.
21. Shilin Pharmaceutical Knowledge Database. <https://www.shilinx.com/>. Accessed 30 Aug 2023.
22. Zhao W X, Zhou K, Li J, et al. A survey of large language models. arXiv preprint. arXiv: 2303.18223. 2023. <https://doi.org/10.48550/arXiv.2303.18223>.

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